SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1707**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

<u>Constraint-Based Problem Solving</u>

Code of Conduct:

I Geferlin Boffy R (192571080) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift

   ii. D2 must work before D3 (Morning < Afternoon < Night)

   iii. Only one doctor per shift

   iv. D3 cannot work Morning

   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles
   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.

c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

ANSWERS:

1. Constraint-Based Problem Solving (Backtracking Search)

Given

Doctors = {D1, D2, D3}

Shifts = {Morning (M), Afternoon (A), Night (N)}

Constraints

1. D1 cannot work Night shift.
2. D2 must work before D3 (Morning < Afternoon < Night).
3. Only one doctor can be assigned to each shift.
4. D3 cannot work Morning shift.
5. Every doctor must be assigned exactly one shift.

Step 1: Assign D1

Possible shifts:

- Morning
- Afternoon
- Night (Not Allowed)

Choose Morning.

Current Assignment:

Doctor	Shift
D1	Morning

Step 2: Assign D2

Remaining shifts:

- Afternoon
- Night

Try Afternoon.

Current Assignment:

Doctor	Shift
D1	Morning
D2	Afternoon

Constraint Check

- D2 is before D3.
- No shift is assigned to more than one doctor.

Proceed to assign D3.

Step 3: Assign D3

Remaining shift:

- Night

Constraint Check

- D3 is not assigned Morning.
- D2 (Afternoon) comes before D3 (Night).
- Each doctor has one shift.
- Each shift has one doctor.

The assignment satisfies all constraints.

Backtracking Search Tree

```
Start
  |
D1 = Morning
  |
  +---- D2 = Afternoon
  |         |
  |         +---- D3 = Night (Solution)
  |
  +---- D2 = Night
            |
            No valid shift available for D3
```

Backtrack

Constraint Verification

Constraint	Result
D1 cannot work Night	Satisfied
D2 before D3	Satisfied
One doctor per shift	Satisfied
D3 cannot work Morning	Satisfied
Every doctor assigned exactly one shift	Satisfied

Final Schedule

Shift	Doctor
Morning	D1
Afternoon	D2
Night	D3

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2. Robot Path Finding using BFS Search

Grid Representation (5 × 5)

S . . X .
. X . X .
. X . . .
. . X X .
X . . . G

Legend

- S = Start
- G = Goal
- X = Obstacle
- . = Free Cell

Manhattan Distance Formula

[
h(n)=|x_1-x_2|+|y_1-y_2|
]

Goal Position = (4,4)

BFS Step-by-Step Node Expansion

Initial Queue

[(0,0)]

Visited Node

(0,0)

Expand (0,0)

Possible Moves

- Right → (0,1)
- Down → (1,0)

Queue

[(0,1), (1,0)]

Expand (0,1)

Possible Move

- Right → (0,2)

Queue

[(1,0), (0,2)]

Expand (1,0)

Possible Move

- Down → (2,0)

Queue
[(0,2), (2,0)]

Expand (0,2)
Possible Move

- Down → (1,2)

Queue
[(2,0), (1,2)]

Expand (2,0)
Possible Move

- Down → (3,0)

Queue
[(1,2), (3,0)]

Expand (1,2)
Possible Move

- Down → (2,2)

Queue
[(3,0), (2,2)]

The BFS process continues in the same manner until the goal node is reached.

Optimal Path
(0,0)
↓
(1,0)
↓
(2,0)
↓
(3,0)
→
(3,1)
↓
(4,1)
→
(4,2)
→
(4,3)
→
(4,4)

Total Cost
Number of Moves = 8
Cost per Move = 1
Total Path Cost = 8

Manhattan Distance Example
At Start Position
$h = |4-0| + |4-0|$
$= 8$
At Position (4,3)
$h = |4-4| + |4-3|$
$= 1$
At Goal Position
$h = 0$

---

3. Autonomous Rescue Robot
(a) Constraint Analysis
Constraint 1: Robot Movement

The robot is allowed to move only in four directions:

- Up
- Down
- Left
- Right

Effect
The robot cannot move diagonally and must plan its path using only these four movements.

Constraint 2: Obstacles
Some rooms contain obstacles.
Effect
The robot cannot pass through blocked cells and must search for an alternative path.

Constraint 3: Limited Sensing
The robot can observe only adjacent rooms.
Effect
The robot does not know the complete environment initially and must gather information while moving.

Constraint 4: Movement Cost

- Normal movement cost = 1
- Risk zone additional cost = 2

Effect
The robot prefers safer routes with lower total cost, even if they are slightly longer.

Constraint 5: Minimize Total Cost
The robot must find a path with minimum overall cost.
Effect
Both travel distance and risky areas are considered while selecting the path.

Constraint 6: Dynamic Knowledge
The robot updates its knowledge whenever new information becomes available.
Effect
If a blocked path or risky zone is discovered, the robot changes its plan and searches for a better route.

(b) Solution Approach
Search Strategy
Uniform Cost Search (UCS) with an Online Search Agent

Steps
Step 1
Start from the initial position (S).
Step 2
Observe the neighbouring cells.
Step 3
Insert all reachable cells into the priority queue according to their cumulative cost.
Step 4
Move to the cell with the lowest total cost.
Step 5
If a new obstacle or risky zone is detected, update the map and calculate a new path.
Step 6
Repeat the process until the robot reaches the goal (G).

Solution Flow
Start
   |
Observe Neighbours
   |
Calculate Cost
   |
Choose Lowest Cost Path

|
Move
|
Update Knowledge
|
Repeat
|
Goal

(c) AI Concepts Used
Intelligent Agent
The rescue robot acts as an intelligent agent because it:

- Collects information using sensors.
- Processes the information.
- Makes decisions.
- Performs actions using actuators.
- Continuously updates its knowledge.

Rationality
A rational agent always chooses the action that minimizes the total cost while reaching the survivor safely.
Example:
Instead of selecting a shorter path with high risk, the robot may choose a slightly longer but safer path to reduce the total cost.

Environment Type

Property	Environment Type
Observability	Partially Observable
Number of Agents	Single Agent
Nature	Dynamic
State	Sequential
State Space	Discrete
Knowledge	Initially Unknown

(d) Justification
Uniform Cost Search with an Online Search Agent is the most suitable approach because:

- It finds the minimum-cost path.
- It considers both movement cost and risk cost.
- It updates the path whenever new information is discovered.
- It works effectively in partially observable environments.
- It helps the robot avoid obstacles and risky zones.
- It ensures safe and efficient navigation until the survivor is reached.
